

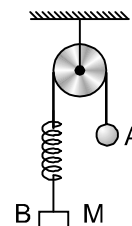
FIITJEE

Work Power and Energy

SECTION – A (Single Correct Answer Type)

This section contains **10 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. In the adjoining diagram, the ball A is released from rest when the spring is at its natural length (neither stretched nor compressed). For the block B of mass M to leave contact with the ground at same time, the minimum mass of A must be
- (A) $M/2$
 (B) M
 (C) $2M$
 (D) A function of M and force constant k of spring

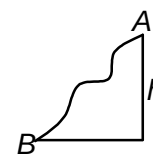


2. A uniform rod of length L and mass M is placed on a rough horizontal surface. The force F applied on the rod is such that the rod is just in the state of rest. If the co-efficient of friction varies according to the relation $\mu = KX$, where K is a positive constant. Suppose friction is present only from $X = 0$ to $X = L$. If the rod moves in the direction of application of force then the total heat generated in the time in which the rod crosses this region, is equal to

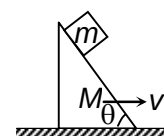


- (A) $\frac{KMgL^2}{g}$ (B) $\frac{KMgL^2}{3}$
 (C) $\frac{2KMgL^2}{3}$ (D) $\frac{KMgL^2}{6}$
3. Two point masses of mass $4m$ and m respectively separated by a distance d are revolving under mutual force of attraction. Ratio of their kinetic energy will be
- (A) $\frac{1}{2}$ (B) 1
 (C) $\frac{3}{4}$ (D) $\frac{1}{4}$

4. When an object is allowed to slide down a hill it stops at the point B because of friction. If friction force depends only on the normal component of the reaction force with coefficient of friction variable along the path, the work done in taking the object slowly from B to A along the hill will be



- (A) mgh (B) $2mgh$
 (C) $> 2mgh$ (D) $< mgh$
5. A block of mass m is stationary with respect to wedge of mass M moving with uniform speed v on horizontal surface. Find the work done by friction force on the block in t second.

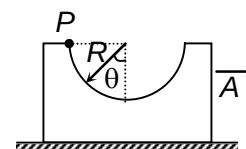


- (A) zero (B) $-\frac{mgvt}{2} \sin 2\theta$

(C) $-\frac{mgvt}{2}$

(D) $-\frac{mgvt}{2} \sin^2 \theta$

6. Inside a smooth hemispherical cavity a particle P can slide freely. The block having this cavity is moving under constant acceleration $A = g \text{ m/s}^2$. A particle is released from the state of rest from the position shown. The angle θ with the vertical, when the particle will have maximum velocity with respect to the block is



(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{3}$

(C) $\frac{\pi}{6}$

(D) $\frac{\pi}{2}$

7. A particle moves on a rough horizontal ground with some initial velocity say v_0 . If $(3/4)^{\text{th}}$ of its kinetic energy is lost due to friction in time t_0 then coefficient of friction between the particle and the ground is

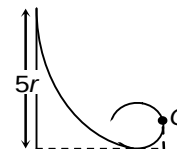
(A) $\frac{v_0}{2gt_0}$

(B) $\frac{v_0}{4gt_0}$

(C) $\frac{3v_0}{4gt_0}$

(D) $\frac{v_0}{gt_0}$

8. An inclined plane ends into a vertical loop of radius r as shown in figure. If a particle of mass m is released from topmost point. The point C on the track as shown in the figure and centre of circular track are at same horizontal level. The normal force at point C is



(A) $8 mg$

(B) $7 mg$

(C) $1 mg$

(D) $3 mg$

9. A block of mass m is placed on another rough block of mass M and both are moving horizontally with same acceleration a due to a force which is applied on the lower block, then work done by lower block on the upper block in moving a distance s will be

(A) Mas

(B) $(m + M)as$

(C) $\frac{M^2}{m}as$

(D) mas

10. The force exerted by a special compression device is given by $F = kx(x - l)$ for $0 \leq x \leq l$, where l is the maximum possible compression and k is a constant. The work required to compress the device a distance d is a maximum when

(A) $d = l$

(B) $d = \frac{l}{2}$

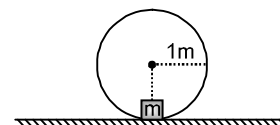
(C) $d = \frac{l}{4}$

(D) $d = \frac{l}{\sqrt{2}}$

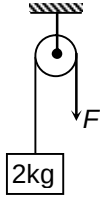
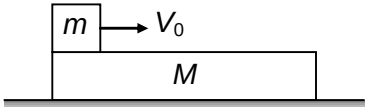
(Multiple Correct Answers Type)

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

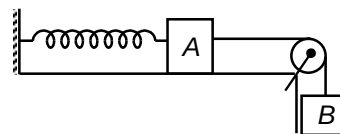
11. A particle of mass m is given an initial speed v inside a smooth fixed spherical shell of radius $R = 1 \text{ m}$ that is just able to complete the circle as shown in the figure



- (A) Acceleration of the particle when its velocity become vertical is $g\sqrt{10}$
 (B) Normal contact force when its velocity become vertical $3mg$

- (C) Speed of the particle when its velocity become vertical is $\sqrt{3g}$
 (D) Contact force at the bottom is $5mg$
12. A particle of mass 5 kg moving in the X - Y plane has its potential energy given by $U = (-7x + 24y)$ Joule. The particle is initially at origin and has velocity $\vec{u} = (14.4\hat{i} + 4.2\hat{j})\text{ m/s}$
 (A) the particle has speed 25 m/s at $t = 4\text{ sec}$
 (B) the particle has an acceleration 5 m/s^2
 (C) the acceleration of particle is normal to its initial velocity
 (D) none of these
13. A ball of mass m is attached to the lower end of a light vertical spring of force constant k . The upper end of the spring is fixed. The ball is released from rest with the spring at its normal (unstretched) length, and comes to rest again after descending through a distance x .
 (A) $x = mg/k$
 (B) $x = 2mg/k$
 (C) the ball will have no acceleration at the position where it has descending through $x/2$.
 (D) the ball will have an upward acceleration equal to g at its lowermost position.
14. A block of mass 2 kg is hanging over a smooth and light pulley through a light string. The other end of string is pulled by a constant force F . The kinetic energy of block increases by 16 J in 2 s , then
 (A) Force F may be 24 N .
 (B) Force F must be 24 N .
 (C) Potential energy must be increased.
 (D) Potential energy may increase.
- 
15. A particle of mass m is attached to a light string of length l , the other end of which is fixed. Initially the string is kept horizontal and the particle is given an upward velocity v . The particle is just able to complete a circle.
 (A) the string becomes slack when the particle reaches its highest point.
 (B) the velocity of the particle becomes zero at the highest point.
 (C) the kinetic energy of the ball in initial position was $\frac{1}{2}mv^2 = mgl$.
 (D) the particle again passes through the initial position.
16. One end of a light spring of spring constant k is fixed to a wall and the other end is tied to a block placed on a smooth horizontal surface. In a displacement, the work done by the spring is $\frac{1}{2}kx^2$. The possible cases are
 (A) the spring was initially compressed by a distance x and was finally its natural length.
 (B) it was initially stretched by a distance x and finally was in its natural length.
 (C) it was initially in its natural length and finally in a compressed position.
 (D) it was initially in its natural length and finally in a stretched position.
17. The coefficient of friction between the block and plank is μ and its value is such that block becomes stationary with respect to plank before it reaches the other end. Then
 (A) the work done by friction on the block is negative
 (B) the work done by friction on the plank is positive
 (C) the net work done by friction is negative
 (D) net work done by the friction is zero
- 

18. In the adjoining figure block A is of mass m and block B is of mass $2m$. The spring has a force constant k . All the surfaces are smooth and the system is released from rest with spring unstretched



- (A) The maximum extension of the spring is $\frac{4mg}{k}$
- (B) The speed of block A when extension in spring is $\frac{2mg}{k}$, is $2g\sqrt{\frac{2m}{3k}}$
- (C) Net acceleration of block B when the extension in the spring is maximum, is $\frac{2}{3}g$.
- (D) Tension in the thread for extension of $\frac{2mg}{k}$ in spring is mg .

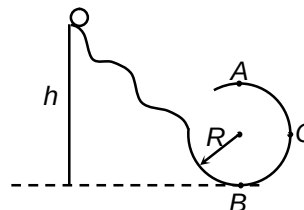
Section – B

Matrix – Match Type

This section contains **4 questions**. Each question has **four statements** (A, B, C and D) given in **Column I** and **five statements** (p, q, r, s and t) in **Column II**. Any given statement in **Column I** can have correct matching with one or more statement(s) in **Column II**. For example, if for a given question, statement B matches with the statements given in q and r, then for that particular question, against statement B, darken the bubbles corresponding to q and r in the ORS.

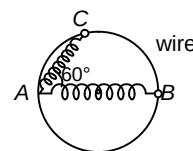
1. A particle is released from height h on a smooth track terminating in a circular path of radius R . A and C are points at top and at horizontal level respectively of the circular path.

Column I represents the different values of height of inclined plane and column II gives the conditions during the motion of the particle.



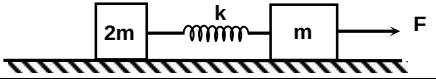
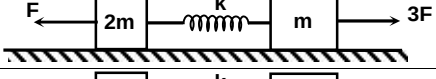
Column I	Column II
A. If $h = 3.2 R$	p. The particle is able to complete vertical circular motion.
B. If $h = 2.7 R$	q. The force exerted by particle on track at point A is zero.
C. If $h = 2.5 R$	r. The force exerted by particle on track at point A is more than its weight.
D. If $h = 4 R$	s. The force exerted by particle on track at point C is more than its weight.

2. A small ring of mass m passes through a smooth wire bent in form of horizontal circle. The ring is connected to a spring whose other end is fixed at A on the wire as shown. The natural length of spring is R and spring constant is $\frac{mg}{R}$ where m is mass of the ring and R is also the radius of circle. Initially the ring is released from rest from position B and it moves towards C as in the figure. (N = Normal reaction between wire and ring, v = speed of ring)



Column-I	Column-II
A. $N = mg$	p. at position B
B. $N = \text{zero}$	q. at position C
C. $N = \sqrt{2} mg$	r. some where between position B and position C.
D. $v = \sqrt{gR}$	s. never

3. In the situations shown in the figure surfaces are frictionless. Find the maximum extension of the springs if blocks are initially at rest and springs are initially in natural lengths.

	Column-I		Column-II
(A)		(p)	$\frac{8F}{3k}$
(B)		(q)	$\frac{14F}{3k}$
(C)		(r)	$\frac{2F}{k}$
(D)		(s)	$\frac{4F}{3k}$
		(t)	$\frac{F}{K}$

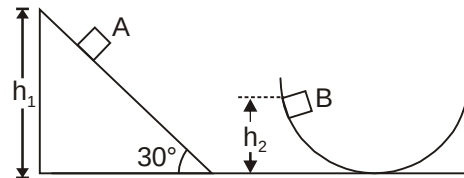
4. A small object of mass 0.5 kg and attached to an end of a massless 2 m long rope is rotated in a vertical circle with the other end of the rope being at the centre of rotation. For any given speed at the lowest point, consider that the motion happens only under gravity, take $g = 10 \text{ m/s}^2$ and match column-I and II:

Column-I	Column-II
(A) If the speed of the object at the lowest point is 3.5 m/s	(p) There will be some point on the circle at which speed of the object is zero but tension in the rope is not zero
(B) If the speed of the object at the lowest point is 8 m/s	(q) There will be some point on the circle for which tension in the rope is zero but speed of the object is not zero
(C) If the maximum tension in the rope is 15 N	(r) The object will not be able to reach the highest point on the circle
(D) If the maximum tension in the rope is 24 N	(s) Linear displacement of the object from the lowest position will be about 5 m.
	(t) Linear displacement of the object from the lowest position will be about 2.8 m.

SECTION – C (Integer Answer Type)

This section contains **8 questions**. The answer to each of the questions is a **single-digit integer**, ranging from 0 to 9. The correct digit below the question number in the ORS is to be bubbled.

1. Two blocks A and B are released on the inclined plane of angle 30° and a circular track of radius R from different height h_1 and h_2 respectively. The mass of each block is m . If F_1 and F_2 are the respective forces experienced by two blocks at the bottom-most point of the tracks and $F_1 = F_2$, then find the value of h_2 (in m) for $R = 8$ m.



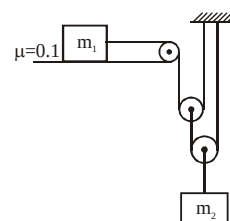
2. Potential energy of a particle moving along x-axis is given by $U = \frac{x^3}{3} - \frac{9x^2}{2} + 20x$. Find out position of stable equilibrium state. (Only conservative forces are working on the particle)
3. A particle of mass m is moving in a circular path of constant radius $r(1\text{m})$ such that its centripetal acceleration a_c is varying with time t as $a_c = k^2 r t^2$, where k is a constant, then power delivered to the particle by the forces acting on it at $t = 5$ sec. (take $mk^2 = 1$ unit)

4. In the figure, find the velocity of m_1 in ms^{-1} when m_2 falls by 9m .

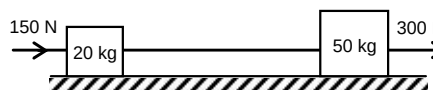
Given $m_1 = m$

$m_2 = 2m$

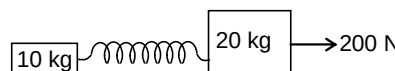
(take $g = 10 \text{ ms}^{-2}$)



5. A block of mass 2.0 kg is pulled up on a smooth incline of angle 30° with the horizontal. If the block moves with an acceleration of 1.0 m/s^2 . The power delivered by the pulling force at a time 4.0 sec after the motion starts is 12 K watts . Find the value of K .
6. Two blocks of masses 20 kg and 50 kg are lying on a horizontal floor (coefficient of friction $\mu = 0.5$) initially string is stretched and blocks are at rest. Now two forces 300 N and 150 N is applied on two blocks as shown in figure. The acceleration of 20 kg block is 2.5 k m/s^2 . Find the value of K .



7. Kinetic energy of a block of mass 2 kg is given as kinetic energy $= 16 t^4$ and the block is moving in a circular path of radius 8 m . Then, find the power generated by the applied force at $t = 1/2$ sec in watt.
8. Two masses of 10 kg and 20 kg respectively are connected by a massless spring as shown in the figure. A force of 200 N acts upon the 20 kg mass. At the instant shown, the 10 kg mass has acceleration of 12 m/s^2 . Find the acceleration of 20 kg mass in m/s^2 .



FIITJEE INTERNAL TEST

ANSWERS KEY

Single correct

- | | | | | |
|-------------|-------------|-------------|-------------|--------------|
| 1. A | 2. B | 3. D | 4. B | 5. B |
| 6. A | 7. A | 8. A | 9. D | 10. A |

Multiple correct

- | | | | | |
|----------------|----------------|----------------|---------------|---------------|
| 11. ABC | 12. ABC | 13. BCD | 14. AD | 15. AD |
| 16. AB | 17. ABC | 18. ABC | | |

Matrix Match

1. **(A) – (p), (r), (s); (B) – (p), (s) ; (C) – (p), (q), (s) ; (D) – (p), (r), (s)**
2. **(A) – (r) ; (B) – (s) ; (C) – (p), (q) ; (D) – (q).**
3. **(A) ì (s) ; (B) ì (s); (C) ì (q); (D) ì (r)**
4. **(A) ì (p, r); (B) ì (q, r); (C) ì (r, t); (D) ì (q, r)**

Integer

- | | | | | |
|-------------|-------------|-------------|-------------|----|
| 1. 2 | 2. 5 | 3. 5 | 4. 4 | 5. |
| 6. | 7. 8 | 8. 4 | | |